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(54) **Title: Toilet-fixture recognition method and system based on a depth camera and a large vision model**

Claims 2 pp.; description 18 pp.; drawings 5 pp.

(57) Abstract

This application discloses a toilet-fixture recognition method and system based on a depth camera and a large vision model, in the field of toilet-fixture recognition. First, the colour image and the depth point cloud are spatio-temporally aligned and fused to obtain high-quality colour-depth aligned data. Then a large vision model extracts features to obtain 2D anchor boxes and class confidence scores, and the 2D anchor box is innovatively used as a pixel-space mask. Next, the class confidence score is used as a registration weighting factor to perform pose registration on the isolated 3D point cloud, preventing the registration from falling into a local optimum and outputting high-precision six-degree-of-freedom pose parameters. Finally, an anti-collision differential check combining the hand-eye calibration matrix and a preset safety-distance threshold computes collision-free working coordinates, closing the technical loop from high-precision multimodal recognition to safe automated robotic-arm operation.

Figure 3 legend: the intelligent cleaning robot (Wangchao C1), labelled parts

1 Depth camera (mounted low on the body)

2 Vision recognition camera (RGB, beside the gripper on the arm end)

3 Magnetic quick-release interface

4 Flexible two-finger gripper

5 6-DoF serial foldable robotic arm

6 High-pressure water-spray end tool

7 Wastewater recovery (suction) end tool

8 Air-drying head end tool

9 Sealed pressure-resistant tubing (routed inside the arm)

10 Sealed clean-water tank

11 Sealed wastewater tank

12 Intelligent electronic control & scheduling unit (built-in AI model library and scheduling engine)

13 Multimodal sensor data real-time cabling

14 Side infrared sensor

15 Multi-line LiDAR system

Other figures

Fig. 1: flow chart of the method (Steps 1-5, same as claim 1). Fig. 2: data-flow diagram (raw sensor stream: restroom colour image + depth point cloud, with intrinsic/extrinsic parameters, into spatio-temporal alignment and fusion, giving aligned colour-depth data. The colour channel goes to the large vision model (multi-layer convolutional features + anchor-box detection), giving 2D anchor-box matrix + class confidence. The 2D box is used as a pixel mask on the depth channel (ROI crop, statistical filtering, clustering), giving an isolated 3D point cloud. With confidence as weighting factor: PCA + ICP pose registration, giving the 6-DoF pose. Hand-eye calibration matrix + safety-distance threshold: world-frame transform + anti-collision check, giving collision-free working coordinates). Fig. 4: Step 3 sub-steps (box corners to binary mask and pixel-level AND crop with hard background deletion; statistical outlier filtering into a smooth cloud; manifold clustering and lossless segmentation from adhering background into the isolated fixture cloud). Fig. 5: system 100 = 110 sensor-data pre-alignment module; 120 semantic extraction / target detection module; 130 3D point-cloud generation module; 140 pose-parameter generation module; 150 working-coordinate generation module.

Claims (machine translation)

1. The toilet appliance identification method based on the depth camera and the visual large model is characterized by comprising the following steps of:

Based on an internal and external reference coordinate system, carrying out space-time alignment fusion on the obtained original sensing data stream to obtain toilet color depth alignment data, wherein the original sensing data stream comprises a toilet color image and a toilet depth point cloud;

Step two, based on a visual large model, carrying out multi-layer convolution semantic feature extraction and anchor frame target detection on color channels in the toilet color depth alignment data to obtain a toilet two-dimensional anchor frame matrix and a toilet category confidence score;

Thirdly, using the toilet two-dimensional anchor frame matrix as a pixel space mask, and performing interested region point cloud clipping, statistical filtering noise reduction and clustering segmentation on depth channels in the toilet color depth alignment data to obtain toilet isolation three-dimensional point cloud;

Step four, using the toilet category confidence score as a registration weighting factor, and carrying out principal component analysis and iterative nearest point pose registration on the toilet isolation three-dimensional point cloud to obtain toilet six-degree-of-freedom pose parameters;

and fifthly, performing world coordinate system transformation and anti-collision differential verification on the six-degree-of-freedom pose parameters of the toilet through the hand-eye calibration homogeneous transformation matrix and a preset safety distance threshold to obtain the non-collision operation coordinates of the toilet.

2. The toilet identification method based on a depth camera and a visual large model according to claim 1, wherein the toilet depth point cloud is collected by the depth camera, and the toilet color image is collected by the visual identification camera.

3. The toilet identification method based on a depth camera and a visual large model according to claim 2, wherein step one comprises:

Based on hardware absolute time stamps carried by the toilet color images and the toilet depth point clouds, performing nearest neighbor time deviation evaluation and out-of-tolerance abnormal frame rejection on the original sensing data stream to obtain synchronous color images and synchronous depth point clouds;

Based on the camera internal parameter matrix and external rigid body conversion parameters, performing space coordinate mapping from a three-dimensional physical rigid body to a two-dimensional pixel plane on three-

dimensional space coordinates of each effective point in the synchronous depth point cloud to obtain a depth pixel projection matrix;

And based on the addressing topological index recorded by the depth pixel projection matrix, performing channel cascade splicing and average filtering vacancy filling on three channel color data of the synchronous color image and the effective depth value of the synchronous depth point cloud to obtain the color depth alignment data between the toilets.

4. The toilet identification method based on a depth camera and a visual large model as claimed in claim 1, wherein the second step comprises:

stripping the color sheets from the color depth alignment data between the toilets and inputting a visual large model to forward propagate multi-scale semantic feature mapping on the color sheets to obtain a high-dimensional multi-scale semantic feature map;

Performing area response activation and candidate anchor frame regression on the high-dimensional multi-scale semantic feature map to obtain an initial generated anchor frame matrix and candidate area feature tensor;

Clustering screening and reserving the overlapped redundant frames in the initially generated anchor frame matrix to obtain a toilet two-dimensional anchor frame matrix;

And carrying out multi-category conditional probability convergence on the candidate regional characteristic tensor to obtain a toilet category confidence score.

5. The toilet identification method based on a depth camera and a visual large model as claimed in claim 1, wherein the third step comprises:

Constructing a corner pixel boundary of a toilet two-dimensional anchor frame matrix as a binarization logic mask, and performing pixel-level logic and projection truncation and background point hard deletion on a depth channel image layer stripped in toilet color depth alignment data to obtain a rough cutting three-dimensional point cloud;

Performing high-frequency outlier statistical filtering noise reduction analysis on the roughly-cut three-dimensional point cloud to obtain a smooth three-dimensional point cloud;

and performing manifold clustering and lossless segmentation on the smooth three-dimensional point cloud and the adhesion background to obtain the toilet isolated three-dimensional point cloud.

6. The toilet identification method based on a depth camera and a visual large model as claimed in claim 1, wherein the fourth step comprises:

Performing geometric centroid and initial axial extraction based on principal component analysis on the toilet isolation three-dimensional point cloud to obtain geometric centroid and principal axis characteristics;

The class labels pointed by the toilet class confidence score are used for calling standard template point clouds from a preset database, and the geometric centroid and the principal axis features are used as initial alignment values to perform confidence weighting iteration closest point registration optimization on the toilet isolated three-dimensional point clouds and the template point clouds so as to obtain an optimal mapping transformation matrix;

And performing homogeneous matrix decomposition and pose parameterization extraction on the optimal mapping transformation matrix to obtain the six-degree-of-freedom pose parameters of the toilet.

7. The toilet identification method based on a depth camera and a visual large model as claimed in claim 1, wherein the fifth step comprises:

Based on the hand-eye calibration homogeneous transformation matrix, performing robot base coordinate system pose mapping on the camera system homogeneous matrix constructed by the six-degree-of-freedom pose parameters of the toilet to obtain world coordinate system pose data;

And performing reverse increment compensation and mechanical arm reachable envelope verification on the position vector of the pose data of the world coordinate system along the normal vector direction of the target surface based on a preset safety distance threshold value so as to obtain a collision-free space operation point

position.

8. The toilet identification method based on the depth camera and the visual large model according to claim 6, wherein the class label pointed by the toilet class confidence score calls a standard template point cloud from a preset database, confidence weighted iterative closest point registration optimization is performed on the toilet isolated three-dimensional point cloud and the template point cloud by using the geometric centroid and the principal axis feature as initial alignment values to obtain an optimal mapping transformation matrix, and the method comprises the following steps:

performing eigenvalue deconstructment of a local neighborhood covariance matrix for each point in the toilet isolation three-dimensional point cloud to obtain local curvature of each point;

Based on the local curvature, the toilet category confidence score and the global registration residual error of each point, constructing a three-factor coupling function to perform nonlinear mapping so as to obtain a self-adaptive dynamic curvature weight value of each point;

And carrying out point-by-point dynamic weight registration optimization on the template point cloud and the toilet isolation three-dimensional point cloud based on the self-adaptive dynamic curvature weight value of each point so as to obtain an optimal mapping transformation matrix.

9. A toilet identification system based on a depth camera and a visual large model, comprising:

The sensing data prealignment module is used for carrying out space-time alignment fusion on the acquired original sensing data stream based on the internal and external reference coordinate system so as to obtain toilet color depth alignment data, wherein the original sensing data stream comprises a toilet color image and a toilet depth point cloud;

The semantic extraction target detection module is used for carrying out multi-layer convolution semantic feature extraction and anchor frame target detection on color channels in the toilet color depth alignment data based on the visual large model so as to obtain a toilet two-dimensional anchor frame matrix and a toilet category confidence score;

the three-dimensional point cloud generating module is used for carrying out interested area point cloud cutting, statistical filtering noise reduction and clustering segmentation on depth channels in the toilet color depth alignment data by taking the toilet two-dimensional anchor frame matrix as a pixel space mask so as to obtain toilet isolation three-dimensional point cloud;

the pose parameter generation module is used for carrying out principal component analysis and iterative nearest point pose registration on the toilet isolation three-dimensional point cloud by taking the toilet class confidence score as a registration weighting factor so as to obtain a toilet six-degree-of-freedom pose parameter;

The operation coordinate generation module is used for carrying out world coordinate system transformation and anti-collision differential verification on the six-degree-of-freedom pose parameters of the toilet through the hand-eye calibration homogeneous transformation matrix and a preset safety distance threshold so as to obtain the non-collision operation coordinate of the toilet.

Description (machine translation)

Toilet appliance identification method and system based on depth camera and vision large model

Technical Field

The application relates to the field of toilet identification, in particular to a toilet identification method and system based on a depth camera and a visual large model.

Background

With rapid development of service robots and automation technologies, automatic cleaning and maintenance operations in complex scenes such as toilets are becoming a focus of attention in the industry. In an automatic operation task, various toilets such as a closestool, a urinal and the like are accurately identified and positioned, and the automatic operation device is a precondition for realizing accurate interaction and operation of a robot or

a mechanical arm. However, the bathroom environment often has challenges such as severe illumination variation, narrow space, high reflectivity of ceramic surfaces, lack of significant textural features, and the like. The three-dimensional space position of the target cannot be obtained only by means of the traditional two-dimensional vision technology, the mechanical arm is difficult to guide to conduct depth operation, and the single three-dimensional point cloud is difficult to accurately analyze semantic information of a complex scene. Therefore, a toilet appliance identification scheme based on the depth camera and the large visual model is constructed, powerful generalized identification and semantic understanding capability of the large visual model are fused with high-precision three-dimensional space perception capability of the depth camera, and the method becomes a necessary development trend for endowing a service robot with comet eyes and realizing high-precision six-degree-of-freedom pose estimation and intelligent operation of the toilet appliance.

However, existing toilet identification schemes based on depth cameras and visual models still have many drawbacks in practical applications. Firstly, in the multi-mode data processing stage, the existing method often lacks a strict space-time alignment and time deviation evaluation mechanism, so that time delay and space projection deviation exist between a color image and a depth point cloud, and high-quality fusion characteristics cannot be formed. Secondly, for the extraction of a toilet target, the detection precision of a conventional visual model is limited when a disordered background is processed, and the two-dimensional semantic anchor frame cannot be effectively utilized as a mask to guide the accurate cutting of the three-dimensional point cloud, so that the extracted three-dimensional point cloud contains a large amount of background noise points and adhesion matters, and the difficulty of subsequent segmentation and calculation is greatly increased. In addition, when three-dimensional pose estimation is performed, the conventional point cloud rigid matching is mostly adopted in the existing algorithm, and the category confidence coefficient of the visual model cannot be fully utilized as the registration weight, so that local optimization is easily involved when the point cloud is absent or has interference, and high-precision six-degree-of-freedom pose parameters cannot be acquired. Finally, most of the existing schemes stay on the pure vision coordinate output level, lack of depth combination with a robot hand-eye calibration system and an anti-collision safety mechanism, and cause that the mechanical arm is extremely easy to collide with and damage the fragile ceramic toilet due to coordinate calculation errors when the mechanical arm executes close operation.

Therefore, a comprehensive toilet identification scheme capable of achieving space-time fusion, accurate denoising, intelligent registration and anti-collision verification is urgently needed in the industry.

Disclosure of Invention

Aiming at the problems in the prior art, the application provides a toilet identification method based on a depth camera and a vision large model, which comprises the steps of firstly, performing space-time alignment fusion on an acquired original sensing data stream based on an internal and external reference coordinate system to obtain toilet color depth alignment data, wherein the original sensing data stream comprises a toilet color image and a toilet depth point cloud, secondly, performing multi-layer convolution semantic feature extraction and anchor frame target detection on a color channel in the toilet color depth alignment data based on the vision large model to obtain a toilet two-dimensional anchor frame matrix and a toilet category confidence score, thirdly, performing interesting area point cloud clipping, statistical filtering and clustering segmentation on the depth channel in the toilet color depth alignment data to obtain toilet isolation three-dimensional point cloud, fourthly, performing main component analysis and iterative closest point gesture registration on the toilet isolation three-dimensional point cloud to obtain a toilet six-degree-of-freedom pose parameter, fifthly, and performing collision-free operation coordinate transformation on the three-dimensional coordinate system through the eye calibration three-dimensional coordinate transformation of the preset three-dimensional anchor frame matrix.

The application further provides a toilet identification system based on the depth camera and the visual large model, which comprises a sensing data pre-alignment module, a pose parameter generation module, a semantic extraction target detection module and a three-dimensional point cloud generation module, wherein the sensing data pre-alignment module is used for carrying out space-time alignment fusion on an acquired original sensing data stream based on an internal parameter coordinate system and an external parameter coordinate system to obtain toilet color depth alignment data, the original sensing data stream comprises a toilet color image and a toilet depth point cloud, the semantic extraction target detection module is used for carrying out multi-layer convolution semantic feature extraction and anchor frame target detection on color channels in the toilet color depth alignment data based on the visual large model to obtain a toilet two-dimensional anchor frame matrix and a toilet category confidence score, the three-dimensional point cloud generation module is used for carrying out interesting region point cloud cutting, statistical filtering and clustering segmentation on the depth channels in the

toilet color depth alignment data to obtain toilet isolation three-dimensional point cloud, the pose parameter generation module is used for carrying out main component analysis and nearest point pose registration on the toilet isolation three-dimensional point cloud based on the visual large model to obtain toilet six-degree-of-freedom coordinate system, and carrying out collision-free coordinate transformation on the three-dimensional coordinate system through the three-dimensional point cloud calibration module and the three-dimensional point cloud calibration module.

Compared with the prior art, the toilet appliance identification method and system based on the depth camera and the vision large model provided by the application are used for solving the problems that in the prior art, multi-mode data fusion quality is poor, point cloud noise points are many, pose estimation accuracy is low, and mechanical arm operation is easy to collide. Firstly, carrying out space-time alignment fusion on the color image and the depth point cloud to obtain high-quality color depth alignment data, and effectively overcoming space-time deviation among multiple sensors. And then, extracting features by using a visual large model to obtain two-dimensional anchor frames and category confidence scores, innovatively taking the two-dimensional anchor frames as pixel space masks, guiding the depth image layer to perform accurate cutting and filtering noise reduction, and realizing efficient stripping of targets and clutter. And then, taking the category confidence score as a registration weighting factor, and carrying out gesture registration on the isolated three-dimensional point cloud to avoid the registration from being trapped into local optimum, thereby outputting high-precision six-degree-of-freedom pose parameters. And finally, combining the hand-eye calibration matrix and a preset safety distance threshold value to perform anti-collision differential verification, calculating a collision-free operation coordinate, and thoroughly opening a technical closed loop from high-precision multi-mode identification to mechanical arm safety automatic operation.

Drawings

The above and other objects, features and advantages of the present application will become more apparent by describing embodiments of the present application in more detail with reference to the attached drawings.

Fig. 1 is a flow chart of a toilet identification method based on a depth camera and a visual large model according to an embodiment of the application.

Fig. 2 is a data flow diagram of a toilet identification method based on a depth camera and a visual large model according to an embodiment of the application.

Fig. 3 is a practical diagram of a structure of an intelligent cleaning robot according to an embodiment of the present application.

Fig. 4 is a flowchart of a third step in a toilet identification method based on a depth camera and a visual large model according to an embodiment of the application.

Fig. 5 is a block diagram of a toilet identification system based on a depth camera and a visual large model according to an embodiment of the application.

Wherein, 1, a depth camera, 2, a visual identification camera, 3, a magnetic quick-release interface structure, 4, a flexible double-finger clamping claw, 5, a six-degree-of-freedom serial foldable mechanical arm, 6, a high-pressure water spraying execution terminal, 7, a sewage recycling execution terminal, 8, an air drying head executing terminal, 9, a closed pressure-resistant pipeline, 10, a sealed water purifying tank, 11, a sealed sewage tank, 12, an intelligent electric control dispatching structure, 13, a multi-mode sensing data real-time transmission wiring, 14, a side infrared sensor, 15 and a multi-line laser radar system.

Detailed Description

Embodiments of the present disclosure will be described in more detail below with reference to the accompanying drawings. It should be understood that the drawings and embodiments of the present disclosure are for illustration purposes only and are not intended to limit the scope of the present disclosure.

Based on the defects in the technical field, the application provides a toilet appliance identification method based on a depth camera and a visual large model. Referring to fig. 1 and 2, fig. 1 is a flowchart of a toilet identification method based on a depth camera and a visual large model according to an embodiment of the present application. Fig. 2 is a data flow diagram of a toilet identification method based on a depth camera and a visual

large model according to an embodiment of the application. In this embodiment, an additional diagram showing a physical execution carrier of the method is introduced, as shown in fig. 3 and related structural descriptions, and fig. 3 is a structural actual diagram of an intelligent cleaning robot according to an embodiment of the present application. The toilet appliance identification method and system based on the depth camera and the vision large model are mainly attached to a multifunctional integrated intelligent cleaning robot hardware platform to realize integral automatic cleaning and safety control for wet closed scenes such as public toilets. The robot main body is provided with the six-degree-of-freedom serial foldable mechanical arm 5 with the space full-posture adjusting capability, the tail end of the robot main body is integrally formed with the magnetic quick-release interface structure 3 and is fixedly connected with the flexible double-finger clamping claw 4, and the robot main body can be quickly replaced and controlled to comprise a high-pressure water spraying execution terminal 6, a sewage recycling execution terminal 7, an air drying head execution terminal 8 and other multifunctional expansion execution terminals. In order to realize the fluid closed loop, the inside of the equipment body is provided with a sealed clean water tank 10 and a sealed sewage tank 11 which are mutually independent, and the sealed clean water tank and the sealed sewage tank directly reach the execution end through a sealed pressure-resistant pipeline 9 which is arranged along the inner cavity of the mechanical arm in a hidden way. On the multidimensional sensing system, the robot platform is provided with a heterogeneous distributed sensor perfectly conforming to the algorithm extraction logic of the application, wherein a visual recognition camera 2 for collecting high-resolution images in real time is fixed beside a clamping claw at the tail end of a mechanical arm so as to follow the mechanical arm to execute the same-frequency approaching and action, a depth camera 1 for sensing macroscopic depth distance and scanning in a large visual field is fixedly and stably arranged at the physical lower end of a machine body, and meanwhile, the machine body is also provided with a side infrared sensor 14 and a multi-line laser radar system 15. The environmental panorama and image features captured by the multi-mode sensing assembly are transmitted back to the intelligent electric control scheduling structure 12 in the body sealed electric control cabin in real time through the built-in multi-mode sensing data real-time transmission line 13, and the software and hardware integrated AI algorithm model library and the intelligent scheduling engine which are carried in the interior comprehensively bear the large model space-time alignment and six-freedom pose optimization calculation provided by the application, so that finally calculated anti-collision three-dimensional safety coordinates are seamlessly converted into physical actions and fluid closed-loop instructions driven by servo control, and the problems of highly complex automatic toilet and dead angle identification cleaning and whole-flow operation tracking are thoroughly solved.

Based on an internal reference coordinate system and an external reference coordinate system, carrying out space-time alignment fusion on the obtained original sensing data stream to obtain toilet color depth alignment data, wherein the original sensing data stream comprises a toilet color image and a toilet depth point cloud. In the application, the depth point cloud of the toilet is collected by a depth camera, and the color image of the toilet is collected by a visual recognition camera. It should be appreciated that in complex working spaces where service robots perform toilet cleaning or maintenance tasks, due to physical space limitations of hardware deployment, a visual recognition camera responsible for acquiring real-time high-resolution detail features is typically mounted on a robotic arm end effector (e.g., beside a gripper jaw) to follow the robotic arm in-phase motion, while a depth camera responsible for wide-area distance perception and large-field spatial profile scanning is fixed to the lower end of the robot body base. The distributed physical space layout structure of the heterogeneous multisensor, together with the natural difference of the sampling frequencies of the hardware at the bottom layers of different sensors, causes the phenomenon of obvious dislocation and cutting of acquired multi-mode data on time and space references. If the misaligned color image and the depth point cloud are directly overlapped or fused, the virtual shadow and edge tearing of the target object can be caused, and serious deviation between the two-dimensional semantic features extracted by the subsequent large vision model and the three-dimensional space position can be caused, so that the mechanical arm is easy to fail in positioning and even crash and damage when executing the operation. In order to eliminate space-time barriers among the multi-source heterogeneous sensors, the first step is introduced, and the necessary precondition for guaranteeing the accuracy of subsequent high-precision semantic extraction and toilet six-degree-of-freedom pose estimation is formed.

In an operable embodiment of the application, the method comprises the steps of carrying out nearest neighbor time deviation evaluation and out-of-tolerance abnormal frame rejection on original sensing data streams based on hardware absolute time stamps carried by a toilet color image and a toilet depth point cloud respectively to obtain a synchronous color image and a synchronous depth point cloud, carrying out space coordinate mapping from a three-dimensional physical rigid body to a two-dimensional pixel plane on three-dimensional space coordinates of each effective point in the synchronous depth point cloud based on a camera internal parameter matrix and external rigid body conversion parameters to obtain a depth pixel projection matrix, and carrying out

channel cascade splicing and average filtering vacancy filling on three-channel color data of the synchronous color image and effective depth values of the synchronous depth point cloud based on addressing topological indexes recorded by the depth pixel projection matrix to obtain toilet color depth alignment data.

The implementation process is as follows, the first stage is nearest neighbor time deviation evaluation and out-of-tolerance abnormal frame rejection. The primary sensor data stream refers to the primary multimedia and depth information stream captured directly by the underlying hardware interface without any secondary processing. After receiving an original sensing data stream transmitted from a communication bus of a bottom device layer such as an industrial control main board, firstly, unpacking the original sensing data stream in a bottom network protocol format, removing a transmission packet header and a check bit, discretizing and stripping the mixed data stream into two independent data threads, wherein one is a high-frequency camera video frame thread, the acquisition frequency of a visual recognition camera is set to be a higher value, for example, 30 frames per second, and the other is a low-frequency depth camera point cloud frame thread, and the frequency of the depth camera is relatively low, for example, 10 frames per second due to the fact that infrared emission and flight time calculation are involved. And then, extracting hardware absolute time stamps carried by all video frames and depth camera point cloud frames after unpacking based on a system global clock crystal oscillator sequence of the intelligent cleaning equipment. The hardware absolute time stamp is directly stamped by the bottom microcontroller at the moment of triggering the sensor hardware exposure, and the absolute physical moment of data acquisition can be accurately reflected. Then, a comparison loop is started by taking a time axis of the high-frequency video frame thread as a reference point, and for each frame in the video frame thread, the absolute values of the time differences between the video frame time stamp and all candidate depth camera point cloud frame time stamps in the current sliding time window are calculated respectively. On the basis, the corresponding point cloud frame serial numbers with the minimum deviation are searched through traversal comparison, so that nearest neighbor matching is realized. In order to ensure the rigor of matching and prevent matching failure caused by network delay or system thread blocking, a preset threshold value is adopted to judge the minimum deviation value. For example, the preset time deviation threshold is set to 15 milliseconds. When the calculated minimum time difference absolute value is smaller than or equal to 15 milliseconds, for example, a certain video frame time stamp is calculated to be one million zero and one hundred and ten milliseconds, the nearest point cloud frame time stamp is calculated to be one million zero and one hundred and twenty two milliseconds, the deviation is 12 milliseconds, the data are judged to be highly consistent in the time dimension and are reserved, if the minimum deviation value is greater than 15 milliseconds due to environmental interference or thread blockage, an abnormal frame rejection mechanism is triggered, and the video frame and the point cloud frame in the time period are judged to be out of tolerance abnormal frame segments and are discarded, so that picture motion blur and space position disjoint caused by time difference in the movement process of the mechanical arm are avoided. And finally, generating a synchronous color image with strict time sequence alignment matching characteristics and a synchronous depth point cloud with consistent time.

After synchronization of the time dimension is completed, a physical mapping phase of the space dimension is entered. Firstly, traversing deconstructing the synchronous depth point cloud, and extracting the three-dimensional geometric space coordinates of each effective point in the cloud set array. The point cloud data captured by the depth camera initially appears as a set of discrete points under the depth camera's own physical coordinate system, containing Cartesian coordinate values of spatial location. In order to convert these three-dimensional coordinates into two-dimensional image views, it is necessary to extract firmware data that is held in the device memory by performing a joint hand-eye calibration step in advance before shipment. The firmware data includes an external rigid body transformation parameter between the camera internal parameter matrix and the depth camera base and the camera origin center. The camera internal parameter matrix describes optical projection properties including focal length, principal point coordinates and the like in the visual recognition camera, and the external rigid body conversion parameters quantify the relative spatial position relationship between the depth camera at the bottom of the machine body and the camera at the tail end of the mechanical arm and consists of a rotation matrix and a translation vector.

For each extracted three-dimensional space coordinate point, performing multi-degree-of-freedom affine transformation and perspective projection operation. Specifically, three-dimensional coordinates of the point cloud are converted from a depth camera coordinate system to a visual recognition camera coordinate system by using external rigid body conversion parameters, and physical geometric points in the sensor three-dimensional coordinate system are precisely flattened and projected onto an imaginary two-dimensional pixel image plane coordinate system by using an internal parameter matrix of the camera. The space coordinate mapping process satisfies the following algebraic geometric relation formula:

In the above-mentioned formula(s), 、 、 Representing three-dimensional real physical coordinates of any effective point extracted from the synchronous depth point cloud under a depth camera world coordinate system, wherein the coordinates of a certain point scanned to the edge of a toilet are 450 mm on the transverse axis, 200mm on the longitudinal axis and 800 mm on the depth axis; representing a3 x 3 orthogonal rotation matrix in the external rigid body transformation parameters, Representing a 3 multiplied by 1 translation vector matrix in external rigid body conversion parameters, and realizing rigid body translation and rotation across a sensor coordinate system under the combined action of the two; Representing an internal parameter matrix of the camera, and consisting of a horizontal focal length, a vertical focal length and an optical center pixel coordinate of the camera; Representing the actual depth value of the point after being converted into the camera coordinate system, namely the distance along the optical axis direction; And Representing the ideal subpixel coordinates of the horizontal and vertical axes on the two-dimensional image plane after projection calculation, respectively. After calculating the coordinates, normalizing the obtained two-dimensional discrete target coordinates, and discarding decimal sub-pixel bits to construct discrete pixel level indexes, e.g. calculating At the end of the line of 452.7, At 805.3, the index is frozen to 452 and 805 after the decimal place is discarded. By constructing a coordinate grid of the two-dimensional pixel level, a topological corresponding relation array of corresponding three-dimensional space point data and two-dimensional pixel coordinates is established, and finally a depth pixel projection matrix containing the strict projection mapping relation is output. The matrix is essentially an addressing topology that records which two-dimensional pixel grid the three-dimensional spatial point should fall into.

Finally, in the channel fusion stage of the data dimension, addressing topology indexing based on depth pixel projection matrix records is performed. First, a high-dimensional tensor of a length-width parameter specification which is completely isomorphic to a synchronous color image is allocated and initialized in a graphics processing memory space of a device as a bearing base, for example, a tensor space with a resolution of 1920×1080 is set. And then, injecting all image pixel data in the synchronous color image into the bearing base according to the original matrix arrangement and channel indexes, so as to form three primary color channels of red, green and blue of the bottom layer, namely a standard color image data layer. And then, based on an accurate addressing topological index link recorded by the input depth pixel projection matrix, the depth values of the effective physical structure provided by the synchronous depth point cloud are correspondingly registered one by one and pushed into corresponding pixel index positions of a fourth dimension space data channel newly added by the bearing base. At this time, a four-channel initial fusion matrix is constructed.

However, in a practical complex bathroom scene, as the toilet is mostly made of smooth ceramic materials, the phenomenon of infrared laser speckle absorption or specular reflection is extremely easy to occur, and the view field angle shielding caused by the difference of the physical positions of the two sensors causes a large amount of depth point clouds to be lost after mapping, so that a fourth dimension (depth channel) presents a plurality of pixel areas without feature mapping such as blank black holes. In order to eliminate the interference of the vacancies on the subsequent vision large model convolution calculation, the outer expansion search filling is executed aiming at the void areas based on the mean filtering interpolation core nearby. The specific operation is to set a sliding neighborhood window with a fixed size, such as a 5×5 pixel matrix window, and slide in the depth channel layer with the blank pixel point as the center. When the depth value of the current pixel is detected to be 0, namely, a gap is detected, a statistical algorithm automatically traverses all adjacent pixels with effective depth values in the 5×5 neighborhood window, sums the effective depth values and calculates an arithmetic average value. Then, the obtained average depth value is used as compensation data to smoothly fill the empty center point position. This process can effectively repair geometric holes due to high light reflection or shielding. The above channel cascade concatenation and mean interpolation Cheng Man are enough to merge the mapping formula:

In the above-mentioned formula(s), Representing the final output and located on the abscissa as And A fused pixel point tensor set containing four channel dimension features; Three channel primary color vectors of red, green and blue representing the bottom layer of the synchronous color image at the coordinate position; Representing an orthogonal concatenation operation performed on the channel dimension of the tensor; Representing a synchronous depth point cloud data array; And Representing the index of the space topology mapping coordinate system analyzed in the depth pixel projection matrix; Then represents a mathematical operator function that performs neighborhood mean filter computation and outer expansion smooth interpolation for the gaps generated after mapping. Through the complex matrix channel splicing and filtering dimension reduction reconstruction, dimension reduction pressing and pasting and solidification packaging of different source head modal data are finally completed, and toilet color depth alignment data with high integration of color

texture information and environmental depth three-dimensional space thickness attribute are generated.

And secondly, based on a visual large model, carrying out multi-layer convolution semantic feature extraction and anchor frame target detection on color channels in the toilet color depth alignment data to obtain a toilet two-dimensional anchor frame matrix and a toilet category confidence score. Accordingly, though the processing of the pre-positioned space-time synchronization pair Ji Yingshe has established uniform multi-mode reference data containing depth space thickness, in the actual toilet cleaning operation space, the background environment is often filled with very disordered visual interference elements such as patterned ceramic tiles, reflecting mirrors, water staggering pipes and the like, and the pure physical depth point cloud lacks advanced semantic information, so that different types of toilet equipment such as toilets, finger basins and the like are difficult to distinguish directly. In order to endow the cleaning robot with advanced visual semantic cognitive ability and further realize accurate isolation and positioning of specific operation targets, the step is led out to provide pixel-level logic mask basis and registration confidence priori conditions for accurate clipping and six-degree-of-freedom pose calculation of the follow-up three-dimensional point cloud.

In an operable embodiment of the application, the method comprises the steps of stripping color slices from toilet color depth alignment data, inputting a visual large model to forward propagate multi-scale semantic feature mapping on the color slices to obtain a high-dimensional multi-scale semantic feature map, performing area response activation and candidate anchor frame regression on the high-dimensional multi-scale semantic feature map to obtain an initial generation anchor frame matrix and a candidate area feature tensor, performing clustering screening and retaining on overlapping redundant frames in the initial generation anchor frame matrix to obtain a toilet two-dimensional anchor frame matrix, and performing multi-category conditional probability convergence on the candidate area feature tensor to obtain a toilet category confidence score.

The implementation is as follows, firstly, a forward propagation multi-scale semantic feature mapping stage is carried out. The algorithm receives the tensor of the toilet color depth alignment data with the length and width specification of 1920 multiplied by 1080 and 4 channel dimensions output by the last link. In the graphics processing unit memory of the device, channel slicing operations are performed to logically decouple these dimensions. Specifically, the 4 th-dimensional depth single channel representing the physical distance is stripped, and a continuous spectrum color patch layer composed of 3 primary color channels of the bottom layer red, green and blue is extracted exclusively, and the tensor shape of the continuous spectrum color patch layer is 1920×1080×3.

And then, inputting the extracted color slice orientation into an airborne artificial intelligence algorithm model library, and activating and loading a visual large model which is trained offline in advance and well-generalized in weight. The visual large model adopts a visual base network based on a self-attention transformation architecture, and the specific architecture comprises an input image blocking embedded layer, a feature encoder formed by alternately stacking a plurality of groups of multi-head self-attention modules and a multilayer convolution backbone network of a bottom layer. The weight, bias and other network parameters of the visual large model are obtained by utilizing a high-definition image training set covering tens of thousands of real toilets and indoor environments in a cloud data center, calculating the cross entropy loss of actual output and manual labeling true values, and performing gradient descent iteration update for hundreds of thousands of times by using a counter-propagation mechanism to obtain the visual large model. At this stage, forward propagation mapping operations are performed on the color patches using the convolution structure and self-attention layer inside the large model. The convolution kernel at the bottom layer is responsible for extracting high-frequency geometric features such as tile joints, equipment edges and the like, and the self-attention encoder at the high layer is responsible for capturing abstract semantic features such as global structured textures of a closestool and curved surface reflection characteristics of a ceramic surface. And finally generating pyramid response features under different resolution scale feature spaces through multistage downsampling and feature aggregation operation, for example, mapping the original 1920×1080 spatial resolution downsampling into multi-dimensional tensors of different scales such as 120×67, 60×34 and the like, and uniformly outputting the multi-dimensional tensors into a high-dimensional multi-scale semantic feature map with the channel depth of 256.

After the rich semantic features are extracted, a stage of performing regional response activation and candidate anchor frame regression on the high-dimensional multi-scale semantic feature map is entered. Based on the generated high-dimensional multi-scale semantic feature map, a large number of preset reference frames with different aspect ratios and basic scales are densely scattered on the coordinate positions of each pixel point of each different feature level. For example, the basic pixel scales are set to 64, 128, and 256, and 3 kinds of horizontal-vertical ratios such as 1 to 1, 1 to 2, and 2 to 1 are given respectively. Then, the region generation

network activator is called for processing. The operator architecture comprises a 3×3 space sliding convolution layer and two parallel 1×1 convolution full-connection branches, and the two kinds of classification judgment and space regression are respectively executed. The algorithm scans and compares pixel gradient peak areas of the high-dimensional multi-scale feature map one by one, and calculates object scores of the inclusion foreground operation targets in each preset reference frame through classification branches, namely, whether the preliminary probability of toilet equipment exists is judged. And (3) intercepting the target pool output of the high-response segment as a candidate region characteristic tensor by setting a score threshold value such as 0.7. Synchronously, for the preliminarily locked object domains to be detected, further calculating nonlinear regression offset of the real boundary of the target relative to the preset reference frame parameters by using regression branches. The calculation process corrects the position boundaries of the candidate boxes by the following spatial regression compensation formula: ; ; ; , wherein, 、 、 、 Respectively representing the horizontal axis coordinate, the vertical axis coordinate, the base width and the base height of the geometric center point of the preset reference frame; 、 、 、 Representing nonlinear regression offset parameters in four degrees of freedom of center bias and scale calculated by the region generating network; 、 、 、 representing the final center coordinates and pixel size of the prediction boundary box generated after compensation correction. By utilizing the set of arithmetic logic, a rectangular pixel coordinate set of a precision envelope potential operation target can be generated, and an initial generated anchor frame matrix with generalized output can be obtained.

The cluster screening reservation phase is performed immediately following the overlapping redundant frames in the initially generated anchor frame matrix. Due to the dense scattering mechanism, for the same toilet equipment within a scene, an initial generation of an anchor frame matrix tends to produce a large number of overlapping redundant frames that are aggregated in the same physical area. After receiving the matrix, a non-maximum suppression algorithm mechanism is applied to the matrix. Firstly, all bounding boxes in the cluster are arranged in descending order according to the calculated gradient response object scores, and the highest score is extracted as a reference inspection box. Then, the space intersection ratio parameter of the residual boundary frame and the reference frame is calculated in sequence, namely the intersection area of the residual boundary frame and the reference frame is divided by the intersection area. And setting an intersection ratio screening threshold value as 0.5, and if the intersection ratio of a certain boundary frame and a reference frame is larger than 0.5, indicating that the boundary frame and the reference frame highly point to the same physical target and the confidence coefficient of the current frame is low, removing the boundary frame from the matrix as a redundant interference item. Through successive iteration loops, only the unique bounding box with the highest gradient response confidence is reserved in each overlapped cluster. For example, the 6 redundant frames originally generated around the toilet are reduced to a single frame, and the two-dimensional coordinate ranges of the redundant frames are precisely locked into the left upper-corner abscissa 420, the ordinate 350 to the right lower-corner abscissa 880 and the ordinate 920. Through the clustering screening and reserving, a clean and accurate final toilet two-dimensional anchor frame matrix is obtained.

And finally, executing a multi-category conditional probability convergence stage for the candidate region feature tensor. And receiving the candidate region characteristic tensor obtained by the preamble in parallel, utilizing a spatial region-of-interest pooling technology to perform downsampling and flattening, and inputting the candidate region characteristic tensor into a full-connection classifier network mounted at the tail of the visual large model. The classifier network is composed of multiple layers of linear mapping matrices for mapping extracted local high-dimensional features to specific classes. In the link, the logistic regression value inside the characteristic tensor is subjected to exponential mapping scaling operation by utilizing a normalized exponential function mechanism. The process aims at setting K preset scene objects within the range, such as setting K equal to 4, respectively representing the categories of a closestool, a urinal, a hand sink, a garbage can and the like, and is used for evaluating independent conditional probability distribution of the objects, wherein convergence calculation meets the following formula: , wherein, Representing the total number of preset scene job target categories to be identified; An original unscaled linear logistic regression score representing the full-connected classifier network output for the ith target class; Representing the exponential operation based on natural logarithmic constant to convert all fractional values into positive numbers and amplify the proportion of high response characteristic, the denominator part is the sum of the exponential scores of all K categories as global normalization constant term; a final independent condition probability value representing that the current candidate region object belongs to the i-th specific physical category, such as toilet. By this formula, a classification judgment confidence probability series composed of floating point numbers in the interval 0 to 1 is obtained, and for example, the probability of belonging to a toilet is calculated to be 0.94, the probability of belonging to a urinal is calculated to be 0.03, and the like. And finally, packaging and outputting the probability

sequence to obtain a complete toilet category confidence score.

And thirdly, using the toilet two-dimensional anchor frame matrix as a pixel space mask, and performing interested region point cloud clipping, statistical filtering noise reduction and clustering segmentation on depth channels in the toilet color depth alignment data to obtain toilet isolation three-dimensional point cloud. It will be appreciated that, through the calculation processing of the preceding steps, although the task object is successfully framed on the two-dimensional image plane through the visual base network and the high-precision class confidence probability is obtained, in the actual physical space of the bathroom, a large number of redundant environmental elements such as background wall bricks, ground structures and water supply pipelines are inevitably attached to the area wrapped by the two-dimensional bounding box. If three-dimensional point cloud data of the whole scene is directly put into a subsequent pose registration link, not only extremely huge calculation force burden is generated, but also serious deviation is generated in a pose matching algorithm due to serious background adhesion noise points, so that real equipment space coordinates cannot be acquired. The method is characterized in that the obtained high-confidence two-dimensional anchor frame is used as a precise pixel space mask, depth channels containing space thickness attributes are precisely cut through cross-modal guidance, free flying points caused by reflection and an environment background tightly adhered in a physical space are stripped layer by layer through high-frequency statistical filtering and three-dimensional manifold clustering technology, and finally, the three-dimensional point cloud of the toilet which is extremely pure and has an independent geometric topological structure is extracted.

Fig. 4 is a flowchart of a third step in a toilet identification method based on a depth camera and a visual large model according to an embodiment of the application. As shown in FIG. 4, in an operational embodiment of the present application, the method comprises a first step of constructing a corner pixel boundary of a toilet two-dimensional anchor frame matrix as a binary logic mask, performing pixel-level logic and projection truncation and background point hard deletion on a depth channel image layer stripped in toilet color depth alignment data to obtain a rough cut three-dimensional point cloud, a second step of performing high-frequency outlier noise statistical filtering noise reduction analysis on the rough cut three-dimensional point cloud to obtain a smooth three-dimensional point cloud, and a third step of performing manifold clustering and adhesion background lossless segmentation on the smooth three-dimensional point cloud to obtain a toilet isolated three-dimensional point cloud.

The implementation process is as follows, firstly, the pixel level logic and projection truncation and background point hard deletion phases are carried out. The control main board receives the toilet color depth alignment data with four-dimensional channels generated in the last processing link, and precisely strips out a fourth-dimensional depth channel layer specially representing the physical three-dimensional absolute space coordinates according to the data registry protocol in the control main board. Then, based on the unique and clean two-dimensional anchor frame matrix received in parallel, the precisely locked identification frame of the toilet is extracted, such as the boundary frame of the locked toilet. Four corner geometric pixel coordinate boundaries of the effective recognition frame are extracted, namely, horizontal extremum and vertical extremum corresponding to the two-dimensional image space, such as a horizontal axis 420 and a vertical axis 350 of the minimum value coordinates of the upper left corner, and a horizontal axis 880 and a vertical axis 920 of the maximum value coordinates of the lower right corner. The corner boundary is constructed into a two-dimensional binarization logic mask matrix with the same resolution in an operation video memory of the equipment, the pixel value inside a mask frame is set to be 1, and the pixel value outside the mask frame is set to be 0.

And then, superposing the logic mask on the surface of the stripped depth channel layer, and performing pixel-level logical AND operation on a pixel-by-pixel basis to perform projection comparison. For the invalid background pixel points outside the calculation boundary mask, as the logic value corresponding to the invalid background pixel points is 0, the algorithm directly executes hard truncation deletion release operation on the depth three-dimensional points indexed by the invalid background pixel points, so that huge and irrelevant background large-environment data are instantaneously removed on a macroscopic level, only valid physical point coordinate space sets which fall in the mask range and are reserved with the logic value of 1 are collected in an aggregation mode, and the valid physical point coordinate space sets are packaged and output to form rough cutting three-dimensional point clouds aiming at a high-probability toilet target area. This process conforms to the following projection truncation logic condition formula:

Wherein, the Representing the rough cutting three-dimensional point cloud set which is preserved through preliminary isolation; representing transverse coordinates corresponding to a two-dimensional grid of pixels And longitudinal coordinates A three-dimensional space effective physical lattice vector at the position; A depth

channel map layer representing the entire spatial field of view stripped from the alignment data; And Representing the minimum value boundary horizontal and vertical pixel coordinates of the upper left corner of the two-dimensional anchor frame; And Representing the maximum value boundary horizontal and vertical pixel coordinates of the right lower corner of the two-dimensional anchor frame; representing logical and boolean operators. The formula severely defines a three-dimensional point cloud that remains only inside the decision window boundary.

And then entering a high-frequency outlier statistical filtering noise reduction analysis stage. And receiving the primarily stripped rough cut three-dimensional point cloud data array, wherein a large number of outlier extremely high frequency free flying points still exist in the point cloud due to multipath reflection of light rays on the ceramic surface of the bathroom or fluctuation of bottom hardware of the photosensitive element. To this end, a multidimensional spatial topology search tree structure based on an approximate nearest neighbor search algorithm is established for the sequence of spatial points in a cache of the device. And traversing each reference coordinate point in the data set, and outwards searching s nearest neighbors in the set three-dimensional physical radius field by using the tree index structure, wherein the number of preset searched neighbor points is 50. The euclidean physical distances of the reference coordinate point to the 50 neighboring points are calculated and the arithmetic mean of the 50 distances is calculated. According to the physical space distribution rule in the nature, the average distance distribution of all coordinate points is considered to accord with a Gaussian normal probability model, and further the expected value of the global distance base large disk, namely the overall average value, and the standard deviation reflecting the discrete degree of the data distribution, namely the fluctuation limit, are obtained through statistical calculation.

Subsequently, the Euclidean distance mean value calculated for each coordinate point is evaluated one by one. When the distance average value of a certain reference point is larger than the set upper limit of the Gaussian statistical threshold value (the upper limit is formed by adding a standard deviation of a set multiple to a global expectation, for example, a multiplier factor is set to be twice the standard deviation), the point is judged to be free from the main structure of the target object, belongs to an interference flying spot, and is thoroughly stripped from the memory. And through comprehensive cleaning and filtering, a smooth three-dimensional point cloud with highly compact surface detail is output. This statistical filtering mechanism follows the following inequality decision formula:

Wherein, the Representing smooth three-dimensional point cloud output after high-frequency flying points are removed; representing a current reference spatial coordinate point being evaluated; representing the reference point and s neighboring points around the reference point The Euclidean distance mean value between the two; Representing the calculated two-norm distance between the three-dimensional vectors; Representing a global average distance expected value obtained by statistics of all point cloud data of the current batch; Represents the global distance standard deviation; A scaling multiplier factor variable representing a predetermined degree of control severity, e.g., set to 2.0. The formula ensures that all the reserved point clouds are closely attached to the surface of a physical entity.

And finally, performing manifold clustering and adhesion background lossless segmentation. And receiving the smooth three-dimensional point cloud from which the free flying points are removed, initializing a blank cluster record table in the running memory of the equipment, and randomly extracting a three-dimensional coordinate point with the attribute which is not marked from the point cloud set as a first cluster growth starting point, namely a seed point. And carrying out neighbor point searching and comparing radially outwards by taking the seed point as a circle center by utilizing the constructed regional space tree topological structure. And when the three-dimensional physical real distance between a certain found adjacent point and the seed center is smaller than a preset critical connectivity threshold size, classifying the adjacent point into the same manifold cluster. The critical connectivity threshold is set strictly according to the real-scene standard process seam drop in the physical world, e.g. there is typically at least 3 cm to 5 cm seam gap between the toilet rim and the rear back wall, bottom tile, so the connectivity threshold is set to 0.04 m, i.e. 40 mm.

Then, the newly incorporated adjacent points are promoted to be new seed centers, the search and judgment are continued outwards in a relay mode, and a communication spreading type growth state is continuously and recursively displayed in the space dimension. When the spread spreads until no new connection point is found within the connectivity threshold radius (40 mm), it is indicated that a physical space fault gap actually existing between the toilet entity and the peripheral wall or the table is encountered in the three-dimensional space, at which time the recursive growth of the current cluster is stopped, and then the above-mentioned process is circulated by re-sowing from the remaining unprocessed loose points. Through comprehensive scanning, the

original point cloud is cut and divided into a plurality of mutually independent geometrical object plaque clusters. And further carrying out volume element space counting comparison aiming at a plurality of plaque clusters generated by stripping. Setting a reasonable point interval, if the number of point clouds of the target toilet is judged to be between twenty thousands points and hundred thousands points, removing too large number of support carrier environment patches such as background wall sections and floor sections containing hundreds of thousands points, and filtering out tiny patches such as water pipe bracket fragments containing only hundreds of points. Through the series of geometric stripping and clearing mechanisms, the toilet isolation three-dimensional point cloud with the independent complete geometric topological shape of the toilet is accurately extracted and output. The mathematical expression of this clustering procedure is as follows:

Wherein, the Representing the pure toilet isolation three-dimensional point cloud of the final filtering output; representing the total number of valid entity plaques meeting the target point size requirement; Performing union set fetching operation on all valid cluster plaque sets meeting the conditions to form a complete entity; representing a central seed node in an activation propagation state in an mth cluster; representing adjacent candidate physical points within the topological space for which connectivity is being retrieved; Representing a critical connectivity space physical distance threshold, e.g., 40 mm as exemplified above, preset in accordance with process specifications. By means of the clustering screening model, the smooth geometric structure of the ceramic curved surface is perfectly reserved, and the adhered physical interferents are subjected to lossless space chopping in a physical sense.

And step four, using the toilet category confidence score as a registration weighting factor, and carrying out principal component analysis and iterative nearest point pose registration on the toilet isolation three-dimensional point cloud to obtain toilet six-degree-of-freedom pose parameters. It should be understood that, through the preamble processing step, although the three-dimensional point cloud of the toilet with extremely pure and independent geometric topology structure has been successfully stripped from the complex toilet environment, and the high confidence probability score that the target device belongs to a specific class is obtained, the three-dimensional point cloud at this time is only a large number of discrete unordered coordinate point sets in the space, and only the geometric profile of the surface of the target object can be represented. For an automatic mechanical arm needing to perform precise lamination and depth interaction, the original point and the rotation direction of a coordinate system which characterize the spatial gesture of a target object are lacking, and an inverse kinematics control instruction cannot be directly generated. In addition, the traditional three-dimensional space registration algorithm generally relies on blind iterative search, is extremely easy to fall into a local optimal solution under the condition of lacking a good initial relative position relation, and fails to integrate semantic confidence of visual model output into space geometric calculation, so that the registration robustness is insufficient. And drawing out the six-degree-of-freedom pose parameters which can directly guide the mechanical arm to execute three-dimensional space interaction operation by using the accurate solution of the step.

In an operable embodiment of the application, the method comprises the steps of carrying out geometric centroid and initial axial extraction based on principal component analysis on toilet isolation three-dimensional point cloud to obtain geometric centroid and principal axis characteristics, calling standard template point cloud from a preset database based on class labels pointed by toilet class confidence score, carrying out confidence weighting iterative closest point registration optimization on the toilet isolation three-dimensional point cloud and the template point cloud by using the geometric centroid and the principal axis characteristics as initial alignment values to obtain an optimal mapping transformation matrix, and carrying out homogeneous matrix decomposition and pose parameterization extraction on the optimal mapping transformation matrix to obtain toilet six-degree-of-freedom pose parameters.

The implementation process is as follows, firstly, the geometric centroid based on principal component analysis and the initial axial extraction stage are carried out. The bottom layer operation unit receives the pure and non-adhesive toilet isolation three-dimensional point cloud output by the last link, and the point cloud set comprises tens of thousands of effective physical points representing the surface profile of a target object such as a toilet. The algorithm firstly carries out traversal accumulation on the coordinates of the point cloud set in the three-dimensional European space, and divides the coordinates by the total number of the point clouds, so that the space center mean value of the geometric body is calculated. For example, after counting fifty thousand effective points, calculating to obtain the geometric center mean value of the toilet point cloud set as 450 mm on the horizontal axis, 200 mm on the vertical axis and 800 mm on the depth axis, wherein the mean value coordinate is taken as the preliminary positioning centroid of the target in the scene. Then, based on the centroid coordinates, a centralized de-averaging process is performed on all the discrete coordinate points in the point cloud set, i.e.,

the original coordinates of each point are subtracted by the centroid coordinates. The algebraic significance of this step is that the origin of the coordinate system of the point cloud is translated to the centroid of the object itself, and the translation bias thereof in the global space is thoroughly eliminated, so that the subsequent gesture analysis is completely independent of the position of the object.

After the centralization is completed, a covariance array matrix reflecting the variance characteristics of the spatial distribution is constructed by utilizing all three-dimensional coordinate vectors after the mean removal through matrix multiplication. The covariance matrix is constructed following the following mathematical formula: , wherein, Representing the calculated 3X 3 dimensional space covariance matrix, which contains the discrete degree information of the point cloud in three orthogonal directions; Representing the total number of effective physical points contained in the toilet isolation three-dimensional point cloud as fifty thousand; representing three-dimensional column vector coordinates of an ith point in the point cloud set; Representing the three-dimensional column vector coordinates of the geometrical centroid of the point cloud set calculated previously Performing a matrix transpose operation on the column vectors on behalf of the column vectors, converting them to row vectors, such that A 3 x 3 local covariance block matrix is generated. And after summing and averaging, carrying out deep feature deconstruction on the covariance matrix by utilizing a singular value decomposition algorithm, and solving the corresponding feature value and feature vector. Among the three orthogonal eigenvectors produced by singular value decomposition, the one with the largest corresponding eigenvalue is extracted. In a physical spatial sense, the feature vector represents the direction in which the distribution of point cloud data is most dispersed and extends longest, i.e. the normal vector corresponding to the geometrical main direction of the target, such as the central axis of fore-aft symmetry or the main tangential plane of the toilet. And finally, carrying out data encapsulation on the calculated three-dimensional centroid coordinates and the main direction vector, and uniformly outputting the three-dimensional centroid coordinates and the main direction vector as geometric centroid and main axis characteristics which roughly represent the pose state of the object.

And then entering a confidence weighting iteration nearest point registration optimization stage. In this processing step, access to a preset database of the device is first required. The preset database is a building information model standard geometric library stored in a solid storage medium in the equipment, and the data of the building information model standard geometric library is derived from a point cloud set converted from a nondestructive grid model directly derived by computer aided design three-dimensional software or through omnibearing scanning of various standard sanitary appliance entities by a high-precision industrial three-dimensional scanner before delivery. Each template point cloud within the database has previously anchored its own coordinate system origin at the geometric center and aligned the principal axis to the standard coordinate axis. The algorithm definitely indicates that the current target is a toilet according to the highest probability class label pointed by the toilet class confidence score transmitted from the last core step, the classification confidence score is 0.94, and the toilet standard template point cloud with the corresponding model is accurately called from the preset database to serve as a registration template set.

And then, using the geometric centroid and the main shaft characteristics calculated in the previous stage as initial values of space transformation, multiplying the initial values by a rotation matrix through one rough rigid body translation, directly moving and preliminarily aligning the adjusted standard template point cloud from the default origin position to the vicinity of the toilet isolation three-dimensional point cloud in the scene, and ensuring that the space postures of the geometric centroid and the main shaft are generally consistent. After the coarse alignment is completed, a confidence weighted iterative closest point optimization loop of the kernel is started. In each calculation iteration, the algorithm searches corresponding point pairs with the nearest space distance based on a multidimensional space tree topological structure between the input target scene point cloud and the standard template point cloud. Unlike conventional algorithms, this link innovatively uses the toilet category confidence score output by the visual large model as a global weight compensation factor for the registration error function. The core objective of the weighted iterative registration process is to minimize the sum of weighted euclidean geometric distances, and the error objective function formula is as follows:

Wherein, the Representing a global registration error scalar that needs to be optimized to a minimum; A toilet category confidence score representing a weight compensation factor, e.g., substituting a specific value of 0.94, amplifying the decision weight of the registration on the overall pose when the confidence is high, and attenuating its sensitivity to fine tuning to prevent divergence when the confidence is low); Representing the total number of valid nearest neighbor matching point pairs found; Representing the coordinate of an ith target point in a toilet isolation three-dimensional point cloud actually captured in a scene; representing the ith model point coordinate matched with the former in the standard template point cloud; representing the 3 x 3 spatial rotation matrix

variables to be solved; Representing the 3 x 1 spatial translation vector variables to be solved. The algorithm solves the optimal rotation matrix by calculating the point-to-cross covariance and using singular value decomposition with continuous iteration And translation vector . When the total error variation calculated between two successive iteration loops is lower than an extremely small set threshold, for example, set to 0.001 mm, it is determined that the point cloud has achieved perfect fitting in a physical sense, and the loop is stopped at this time. Rotation matrix obtained by final convergence And translation vector And (3) performing matrix reorganization, and outputting an optimal mapping transformation matrix containing 4×4 dimensions, wherein the matrix completely records the whole rigid transformation process from the standard posture to the current real physical posture.

In particular, in a standard spatial pose iterative reconstruction process, if only the toilet class confidence score output by the visual large model is directly applied as a static, global linear scalar weight to the Iterative Closest Point (ICP) registration energy function, a series of significant technical bottlenecks may be introduced. Since a single scalar is used to apply exactly the same weight to all pairs of registration points, this global uniformity results in an algorithm that cannot distinguish between geometric reliability differences in different physical areas in the three-dimensional point cloud, e.g., geometric distinctions of contour feature points such as toilet edges, tank corners, etc., and registration constraints are much higher than smooth featureless ceramic sidewall surface points, but in traditional linear registration both are treated incorrectly. In addition, because the mapping relationship between the visual confidence score and the spatial registration accuracy is not a simple linear relationship, direct use can lead to the problem of linear response mismatch, namely, when the confidence is in the middle fuzzy interval from zero five to zero seven, the linear weight cannot generate enough attenuation to effectively inhibit registration coordinate drift caused by wrong template matching, and when the confidence is extremely high, for example, more than 0.95, the linear weight lacks an exponential amplification mechanism, and the high confidence cannot be fully utilized to accelerate the convergence of the pose matrix. More importantly, if the weight factor is kept constant in the whole hundred registration iteration loops, dynamic self-adaptive adjustment cannot be performed along with gradual decline of the registration residual error, the weight factor must have identical constraint strength in the initial registration stage (wide search is needed to avoid the locally optimal coarse alignment stage) and the final registration stage (strict constraint is needed to be applied to solve the extremely high precision fine convergence stage), so that the search of the globally optimal spatial solution is seriously interfered. In order to thoroughly break through the limitation of the linear and static weights, the step is led out and used for constructing a nonlinear optimization mechanism capable of simultaneously sensing semantic confidence, local geometric curvature and global time sequence residual error states in a sharp mode, and reasonable constraint force is dynamically distributed for each physical space point in an iteration process so as to greatly improve the robustness and absolute precision of the finally solved mechanical arm operation pose parameters.

The specific implementation process of the nonlinear weight iteration mechanism based on point-by-point self-adaption and dynamic evolution along with iteration covers three continuous precise calculation stages of local curvature deconstruction, three-factor coupling mapping and dynamic weighting registration.

Based on this, in a preferred embodiment of the present application, retrieving a standard template point cloud from a preset database based on a class label pointed to by a toilet class confidence score, performing confidence weighted iterative closest point registration optimization on a toilet isolated three-dimensional point cloud and a template point cloud by using a geometric centroid and a principal axis feature as initial alignment values to obtain an optimal mapping transformation matrix, including:

And performing eigenvalue deconstructment of the local neighborhood covariance matrix for each point in the toilet isolation three-dimensional point cloud to obtain the local curvature of each point. It should be appreciated that the higher alignment priority of feature salient regions such as edges, corners can be given in the subsequent registration only if the geometrical saliency of the microscopic region where each three-dimensional coordinate point is located is precisely quantified. In specific implementation, the toilet isolation three-dimensional point cloud containing tens of thousands of effective physical coordinates is received, such as toilet point cloud containing fifty thousands of effective points extracted at the front stage. And multiplexing a microcosmic principal component analysis framework constructed in advance aiming at each independent point in the set, and extracting an adjacent point set which takes the point as a center and is in a preset physical radius. Specifically, a multidimensional space topology search tree structure pre-built in a memory is called, a three-dimensional physical point currently being evaluated is taken as a sphere center, a physical radius is searched in a preset micro space, for example, the radius is set to be 15 mm, and the search is performed outwards in a sphere

divergence mode, so that a local adjacent coordinate point subset wrapped in the micro spherical envelope line is accurately captured, for example, 35 closely connected adjacent points are searched at the edge of a toilet. And then, carrying out traversal accumulation on three-dimensional coordinate column vectors of all adjacent points in the microcosmic subset by an algorithm, dividing the three-dimensional coordinate column vectors by the number of points, and solving the geometric center mean value coordinate of the local micro tangential plane region. After the local center is established, the absolute coordinates of each point in the subset are respectively subtracted by the center mean value coordinates to perform local de-averaging treatment, so that the origin of the local coordinate system is translated to the centroid of the micro-section. And finally, multiplying the relative space three-dimensional column vectors subjected to the local de-averaging treatment by the outer product matrix of the relative space three-dimensional column vectors and the transposed row vectors of the relative space three-dimensional column vectors, accumulating all generated matrices in the subset, and then averaging, so that a covariance matrix aiming at the local neighborhood is strictly constructed in a mathematical physical layer. Through the set of precise cutting processes of point-by-point scanning and microcosmic dimension reduction, the algorithm can convert abstract object surface relief into a tiny local feature distribution form which can be analyzed by a quantization matrix. And then, performing singular value decomposition operation on the covariance matrix, and solving three eigenvalues reflecting the distribution characteristics of the local space point cloud. Based on the three eigenvalues, the local curvature of the point is calculated through specific algebraic ratio operation. The calculation process follows the following local curvature characteristic formula:

In the above-mentioned formula(s), A local surface curvature scalar value representing the calculated i-th three-dimensional point; 、 And Three eigenvalues obtained by deconstructing the local neighborhood covariance matrix which together represents the ith point The value of the three features is the one with the smallest value, and the physical meaning of the value directly corresponds to the spatial discrete degree or thickness of the local curved surface along the normal vector direction. For example, when the scan point is located at a high curvature profile of the toilet rim, the eigenvalues may be resolved to 10.5, 8.2, and 4.1, and substituted into the formula to calculate the local curvature About 0.18, and the characteristic values may be 12.0, 11.5 and 0.2 when the scan point is on a flat surface of the toilet tank, the calculated local curvature is only 0.008. The abstract geometric surface relief is successfully converted into an accurate quantization index which can be directly called by an algorithm, and a data base of physical geometric dimension is laid for breaking global uniformity weight.

Based on the local curvature of each point, the toilet category confidence score and the global registration residual error, a three-factor coupling function is constructed to carry out nonlinear mapping so as to obtain the self-adaptive dynamic curvature weight value of each point. Correspondingly, the information of a single dimension is insufficient to cope with complex field reflection and shielding interference, a nonlinear mathematical model is required to be established, and macroscopic semantics, microscopic geometrical curvature and macroscopic time sequence states of a registration process provided by a visual large model are subjected to depth fusion, so that intelligent distribution of weights is realized. In specific implementation, the algorithm receives the confidence scores of the toilet category output in the visual stage in parallel, such as 0.94, the local curvature calculated in the last link, and the registration residual state recorded in the system memory. On the basis, a three-factor coupling function comprising a Sigmoid family nonlinear mapping term, a geometric enhancement term and a time sequence attenuation term is constructed. The algebraic expression of this function is as follows:

in the above-described coupling weight formula, Representing an adaptive dynamic curvature weight value which is calculated and generated specially aiming at an ith registration point pair in an nth registration iteration loop, wherein the first term of the formula is a Sigmoid nonlinear semantic mapping term An original value representing the entered toilet category confidence score such as 0.94, Represents a preset confidence decision threshold center point, is set to 0.6, Representing a sharpness control parameter, e.g. preset to 10.0, for adjusting the width of the sensitive interval of the nonlinear mapping, the second term of the formula being a geometric enhancement term, wherein In order to introduce a local curvature into the body, For the preset curvature enhancement coefficient, for example, 5.0 is set for controlling the contribution amplitude of the geometric feature to the weight, and the third term of the formula is a time sequence residual error attenuation term Representing the global registration residual energy value at the end of the n-1 th iteration, Representing the initial global residual energy value at the moment of the algorithm starting, and the ratio of the initial global residual energy value and the initial global residual energy value forms a normalized time sequence evolution reference. The global time sequence attenuation ratio applies the same scaling coefficient to all registration point pairs in the same round of iteration, the core effect of the global time sequence attenuation ratio does not change the relative constraint priority among points, namely the differential constraint force among points is completely determined

by a local curvature enhancement item independently calculated point by point in the first two items, but dynamically normalizes and regulates the absolute magnitude of a registration energy function at an engineering numerical value calculation level, namely when the global residual error is larger at the initial stage of registration, the ratio is close to 1.0 so as to maintain the energy function in a higher magnitude range, each matrix element has sufficient effective digital precision when the weighted cross covariance matrix is solved by singular value decomposition, the ratio synchronously attenuates so as to compress the absolute value of the energy function to a reasonable range matched with floating point operation precision along with gradual convergence of iteration advancing residual error, thereby effectively preventing the condition number degradation of the covariance matrix caused by excessively small residual error, the accumulated amplification of floating point truncated error, the stability of the numerical value oscillation of a solver and the like at the engineering implementation level, and further ensuring the robustness stability of numerical value solution in the whole iteration optimizing process in tens of hundreds of times or even hundreds of cycles. To be at the high curvature point of the toilet rim, For example, 0.18, because the confidence level is 0.94 and is far higher than the threshold 0.6, the calculated value of the sigmoid term approaches 0.967, the geometric enhancement term is amplified to 1.9, if the residual ratio is reduced to 0.2 at the tenth iteration, the final coupling weight of the point is about 0.367, and because the local curvature is only 0.008, the geometric enhancement term is only 1.04, and under the action of the same global residual ratio of 0.2, the weight of the point is only 0.201. It can be seen that the weight difference of about 1.83 times between the high curvature edge point and the flat surface point is completely derived from the synergistic effect of the point-by-point independent Sigmoid semantic mapping and the local curvature geometric enhancement in the first two items, and the relative weight proportion of the difference cannot be counteracted by the global equal-ratio scaling of the third item in the singular value decomposition solving process, so that the solver can be ensured to be preferentially aligned with the key contour feature points with high geometric distinguishability in each round of iteration. In this way, noise is strongly restrained in a fuzzy region through Sigmoid nonlinear mapping, weights are fully released in a confidence region, and the greater registration contribution given by high curvature points is utilized, so that the geometric precision of point-by-point differentiation constraint and the numerical calculation stability of a long-range iteration process are perfectly considered in combination with dynamic normalization stable regulation and control of the global residual error attenuation item on the numerical magnitude of an energy function.

And carrying out point-by-point dynamic weight registration optimization on the template point cloud and the toilet isolation three-dimensional point cloud based on the self-adaptive dynamic curvature weight value of each point so as to obtain an optimal mapping transformation matrix. That is, the differential weight calculated by the front link fine center is required to be truly applied to the physical solving process of space alignment, so as to guide matrix transformation to approach towards the direction with most obvious geometric features and most reliable semantics. In specific implementation, the algorithm invokes the standard template point cloud of the closestool with the corresponding model extracted from the preset database, and takes the initial geometric centroid and the main shaft as initial values of space transformation. In the iteration solving loop, the generated point-by-point dynamic weight is utilized to thoroughly replace the global static scalar in the traditional ICP energy function, and the Euclidean distance square between the input actual observation point and the template reference point is subjected to differential weighted summation. The energy objective function formula of the weighted registration optimization process is as follows:

Wherein, the Representing a weighted registration residual energy total value which needs to be minimized by an algorithm core in an nth iteration loop; representing the total number of valid registration point pairs found in the run; representing the self-adaptive dynamic curvature weight value specific to the ith point pair substituted in the round; representing an ith space observation point column vector in the toilet isolation three-dimensional point cloud from real extraction in the scene; Representing an ith reference point column vector matched with the former in the standard template point cloud; representing three rows and three columns of space rotation matrixes to be solved and optimized; representing three rows and one column of space translation vectors to be solved; The square of the Euclidean distance in three-dimensional space is obtained. The operation engine is continuously adjusted And The overall energy function value is made to approach a minimum until the energy difference between the two iterations is less than a predetermined convergence minimum, such as 0.001 mm. Since the high-recognition toilet edge points take the dominant weight in the summation formula, the solver will preferentially and precisely align these critical geometric frames, allowing reasonable tolerances in the flat areas. This greatly improves the robustness of the final output optimal mapping transformation matrix, and practice shows that this mechanism can reduce the small calculation error in the rotation matrix by about thirty percent to fifty percent. The leap of pose accuracy directly reduces the volume of an uncertainty envelope body in a subsequent anti-collision safety verification stage, so that the mechanical arm can safely and rapidly execute

flushing and brushing actions at an absolute spatial position extremely close to the surface of the fragile ceramic, and the coverage rate and engineering efficiency of automatic cleaning operation are thoroughly improved.

And finally, performing homogeneous matrix decomposition and pose parameterization extraction. The algorithm receives the homogeneous optimal mapping transformation matrix with four rows and four columns in dimension, and in order to enable the homogeneous optimal mapping transformation matrix to be recognized and called by a motion control algorithm of a mechanical arm bottom layer, the homogeneous optimal mapping transformation matrix is deconstructed into a parameter form with more physical significance. First, orthogonal rotation components (i.e., rotation matrices) corresponding to the first three rows and the first three columns and coordinate origin translation components corresponding to the fourth three rows and the first three columns are split and stripped therefrom. And directly extracting three numerical values of an abscissa, an ordinate and a depth coordinate in the translation component as an absolute description of the spatial position of the target, wherein the extracted spatial position is 452.3 mm on the horizontal axis, 198.6 mm on the vertical axis and 803.1 mm on the depth axis. Based on classical geometric triangle transformation theory, the extracted rotation component is converted into Euler angle representation form which is more in line with inverse kinematics calculation solution of the mechanical arm, namely yaw angle, pitch angle and roll angle. The rotation matrix to euler angle nonlinear mapping solution process follows the following continuous inverse trigonometric function formula: ; ; . Wherein the variables are 、 、 、 、 Representing specific numerical elements of corresponding rows and columns, respectively, in the split 3 x 3 rotation matrix, e.g Elements representing the first column of the third row of the matrix); Representing a bivariate arctangent mathematical function, which can accurately judge the specific quadrant where the calculated angle is located according to the signs of two input parameters, so as to avoid the ambiguity of positive and negative spans generated by the traditional arctangent function; representing the calculated pitch attitude rotated about the vertical longitudinal axis; Representing a yaw angle attitude rotated about a depth optical axis; Representing the roll angle attitude of rotation about the horizontal transverse axis. Three specific radian values representing the rotation gesture are obtained through strict triangular operation, and if the rolling angle is calculated to be 0.05 radian, the pitch angle is calculated to be-0.12 radian and the yaw angle is calculated to be 0.86 radian. Finally, the extracted three-dimensional absolute position coordinate signal and the calculated three-dimensional Euler angle posture signal are subjected to high-level fusion packaging to form an array containing position and direction full-dimension information. The array is finally output as the toilet six-degree-of-freedom pose parameter with the potential of the space full-pose adjustment instruction.

And fifthly, performing world coordinate system transformation and anti-collision differential verification on the six-degree-of-freedom pose parameters of the toilet through the hand-eye calibration homogeneous transformation matrix and a preset safety distance threshold to obtain the non-collision operation coordinates of the toilet. That is, although the six-degree-of-freedom pose parameters of the toilet with extremely high precision have been accurately calculated through the complex calculation process of the preamble, it should be noted that the set of pose parameters is only the relative position and pose relationship established under the local optical coordinate system inside the depth camera. For an automated cleaning robot responsible for performing the final actions, the underlying motion planning instructions of its robotic arm are highly dependent on a global world coordinate system established on the physical reference origin of the robot chassis. The robotic arm will not recognize the position command without uniformly translating the coordinate system. In addition, ceramic sanitary wares such as toilets, finger bowls, and the like are fragile and scratchable items. If the tail end of the mechanical arm is directly guided to be closely attached to the absolute coordinate of the physical surface of the target after conversion, rigid impact is easily generated due to the motion inertia of the mechanical arm or the execution error of the motor, so that the equipment is damaged. The step of drawing out and using is to accurately map the local visual coordinates into a global motion coordinate system of the robot, and to carry out physical-level anti-collision reverse compensation and kinematic envelope verification through fusion process safety distance, so that a technical closed loop from multi-mode visual perception positioning to safe, accurate and automatic physical close operation is thoroughly opened, and a group of working coordinates which can directly drive equipment and are absolutely collision-free are ensured to be finally output.

In an operable embodiment of the application, the fifth step comprises the steps of performing robot-based coordinate system pose mapping on a camera system homogeneous matrix constructed by six-degree-of-freedom pose parameters of a toilet based on a hand-eye calibration homogeneous transformation matrix to obtain world coordinate system pose data, and performing inverse increment compensation on a position vector of the world coordinate system pose data along a normal vector direction of a target surface based on a preset safety distance threshold and performing mechanical arm reachable envelope verification to obtain a collision-

free space operation point.

The implementation process is as follows, firstly, a robot base coordinate system pose mapping stage is carried out. The bottom master control operation unit receives the six-degree-of-freedom pose parameters of the toilet, which are output by the front link and contain the absolute azimuth characteristics of the target. The parameter internally comprises a three-dimensional Euler angle posture array and a three-dimensional position coordinate vector. According to the principle of three-dimensional space rigid body kinematics, firstly, the extracted Euler angle components are reconstructed into three rows and three columns of standard rotation matrixes with orthogonal characteristics through internal inverse trigonometric function matrix reduction operation of the rolling angle 0.05 radian, the pitch angle-0.12 radian and the yaw angle 0.86 radian calculated in the previous steps. Meanwhile, the extracted spatial position coordinates, such as the transverse axis 452.3 mm, the vertical axis-198.6 mm, and the depth axis 803.1 mm, are taken as spatial translation vectors. Then, the 3×3 rotation matrix is embedded into the upper left corner of a four-row four-column blank matrix, the translation vector is embedded into the first three rows of the fourth column of the matrix, and three 0 s and one 1 s are complemented in the last row of the matrix, so that a four-by-four-dimensional camera system homogeneous matrix is formed by combining the two.

And then extracting the hand-eye calibration homogeneous transformation matrix which is burnt in advance from the bottom layer parameter configuration solid-state storage module. The hand-eye calibration homogeneous transformation matrix is also a four-row four-column high-dimensional rigid transformation matrix, and has the core function of representing the fixed geometric relative relation between the optical axis center inside the vision sensor and the world physical coordinate system (namely a base coordinate system) where the robot reference chassis is located. The matrix is obtained by utilizing a standard black-and-white checkerboard calibration plate to control a mechanical arm to execute high-frequency visual shooting under a plurality of known kinematic postures in the factory assembly or the primary calibration stage of the robot equipment, and further accurately solving, solidifying and storing each joint deviation through a plurality of groups of simultaneous equations based on dual quaternion or nonlinear optimization. After the bottom calibration matrix is obtained, the operation control unit strictly utilizes a matrix multiplication rule in linear algebra to multiply the camera system homogeneous matrix representing the relative position of the target by the hand-eye calibration homogeneous transformation matrix, so that conversion projection of the target object from a virtual visual optical space to a real physical operation space which can be understood by the mechanical arm is realized in mathematical dimension. The mathematical mapping expression of the core coordinate system transformation process is as follows: , wherein, Representing a four-by-four camera system homogeneous transformation matrix which is formed by reconstructing and splicing six-degree-of-freedom pose parameters of a preamble toilet, and is completely established under a local optical coordinate system of a depth sensor; Representing a high-precision hand-eye calibration homogeneous transformation matrix which is extracted from a bottom layer persistence configuration file, and containing physical installation offset compensation information of a sensor; Representing the four-by-four-dimensional homogeneous transformation matrix which is output after the operation is completed and is completely mapped on the reference origin of the robot global chassis. After completing the matrix multiplication mapping, from the And re-stripping out the converted brand-new rotation matrix component and translation position component from the matrix, and re-packaging the rotation matrix component and the translation position component into world coordinate system pose data. The generated data, such as the converted global world coordinates, is changed to 810.2 mm on the horizontal axis, 50.6 mm on the vertical axis and 605.4 mm on the vertical axis, can be directly used for driving the bottom layer guidance of the servo motor, and has extremely high physical execution value.

And then entering a reverse increment compensation and mechanical arm reachable envelope curve verification stage. And after the pose data of the world coordinate system are acquired, extracting a three-dimensional space absolute coordinate vector and a rotation matrix representing the space orientation of the target. In practical automated toilet cleaning operations, if the robot end effector is directly controlled to move to the absolute coordinate point of the physical surface, the effector itself inevitably collides with the ceramic surface rigidly. In order to avoid the risk, a built-in physical obstacle avoidance and tool parameter library is called in real time, and preset safety distance thresholds corresponding to different cleaning tool heads such as a high-pressure wide-width water spray head, a sewage vacuum suction head or a soft hair rotary brush head mounted at the tail end of the current mechanical arm are extracted. The preset safety distance threshold is a scalar length limiting parameter with a definite physical anti-collision meaning that defines the minimum allowable spatial physical separation that must be maintained between the end hardware device of the mechanical arm and the ceramic surface of the toilet to maintain normal spraying or brushing operation. The threshold is comprehensively set by

testing a large number of optimal working hydrodynamic distances of the cleaning tool before delivery and superposing and considering the maximum brake overshoot error distance of each joint servo motor. For example, when the mounting tool is currently identified as a high-pressure wide sprinkler head, a preset safe distance threshold is dynamically extracted and set to 150 mm in order to ensure that the water flow fan just covers the inner wall of the toilet bowl and to absolutely avoid the metal sprinkler head touching the ceramic rim. The algorithm then extracts column vector data corresponding to the depth axis direction from the rotation matrix of pose data, the unit vector representing physically geometrically the normal vector direction of the target object (e.g., a contaminated surface of a toilet), i.e., the spatial forward direction perpendicular to the target tangential plane. Then, on the basis of the position vector coordinates of the world coordinate system pose data, the algorithm instructions perform an inverse incremental compensation operation on the target along the normal vector direction. In popular terms, the working point of the tail end of the mechanical arm in planning is forced to retract backwards by a safe distance along a track perpendicular to the surface of the toilet. The space reverse collision prevention backoff compensation operation accords with the following three-dimensional vector compensation formula:

Wherein, the Representing an absolute physical three-dimensional coordinate column vector of the high-precision target surface extracted from pose data of a world coordinate system; A preset safety distance threshold representing the acquired scalar form, namely a previously specifically set value of 150 mm; Representing a three-dimensional unit normal vector of the target surface stripped from the gesture rotation matrix, representing a positive vertical axis of the interactive interface; representing the hover point position coordinates of the preliminary safe residence space calculated after the reverse increment compensation operation.

After the preliminary security coordinates are calculated, the underlying motion planning layer must also perform a rigorous differential verification logic. The verification logic inputs the compensated space point position coordinates to the mechanical arm inverse kinematics solver module for simulation operation. On one hand, whether the torsion angle of each mechanical joint required for verifying the point position coordinates and the corresponding gesture exceeds the physical reachable working envelope boundary of the mechanical arm, and the phenomenon that singular points or dead angles cause jamming is prevented. On the other hand, the established static toilet environment collision box model is called, and whether the geometric sweep track of the connecting rod to be generated collides with the space interference of inherent environment barriers (such as glass partition plates, side water supply pipes and other non-traversable areas of the toilet) preset in the space is checked. If collision early warning occurs, the algorithm sets step fine adjustment compensation angles around the normal line, and substitutes the step fine adjustment compensation angles again for verification until no interference passes. After the series of collision prevention compensation and inverse kinematics differential verification screening based on strict physical boundaries, each parameter is finally locked and packaged, and a group of toilet collision-free operation coordinates which have absolute space coordinate positioning information and full-posture inclination angle guiding information and are subjected to collision immune authentication are output. At this time, the bottom servo driver generates a smooth motion level pulse according to the final operation coordinates, and controls the mechanical arm to safely, rapidly and accurately expand the automatic maintenance work against the target area.

In summary, the toilet identification method based on the depth camera and the large vision model according to the embodiment of the application is clarified, so as to solve the problems of poor multi-mode data fusion quality, more point cloud noise points, low pose estimation precision, easy collision of mechanical arm operation and the like in the prior art. Firstly, carrying out space-time alignment fusion on the color image and the depth point cloud to obtain high-quality color depth alignment data, and effectively overcoming space-time deviation among multiple sensors. And then, extracting features by using a visual large model to obtain two-dimensional anchor frames and category confidence scores, innovatively taking the two-dimensional anchor frames as pixel space masks, guiding the depth image layer to perform accurate cutting and filtering noise reduction, and realizing efficient stripping of targets and clutter. And then, taking the category confidence score as a registration weighting factor, and carrying out gesture registration on the isolated three-dimensional point cloud to avoid the registration from being trapped into local optimum, thereby outputting high-precision six-degree-of-freedom pose parameters. And finally, combining the hand-eye calibration matrix and a preset safety distance threshold value to perform anti-collision differential verification, calculating a collision-free operation coordinate, and thoroughly opening a technical closed loop from high-precision multi-mode identification to mechanical arm safety automatic operation.

Fig. 5 is a block diagram of a toilet identification system based on a depth camera and a visual large model according to an embodiment of the application. As shown in FIG. 5, a toilet identification system 100 based on a depth camera and a visual large model according to an embodiment of the application comprises a sensing data pre-alignment module 110 for performing space-time alignment fusion on an acquired original sensing data stream based on an internal and external reference coordinate system to obtain toilet color depth alignment data, wherein the original sensing data stream comprises a toilet color image and a toilet depth point cloud, a semantic extraction target detection module 120 for performing multi-layer convolution semantic feature extraction and anchor frame target detection on color channels in the toilet color depth alignment data based on the visual large model to obtain a toilet two-dimensional anchor frame matrix and a toilet category confidence score, a three-dimensional point cloud generation module 130 for performing point cloud clipping, statistical filtering and noise reduction and clustering segmentation on the depth channels in the toilet color depth alignment data to obtain a toilet isolation three-dimensional point cloud, a pose parameter generation module 140 for performing main component analysis and nearest point pose registration on the toilet isolation three-dimensional point cloud based on the visual large model, and performing six-dimensional coordinate transformation on the three-dimensional point cloud as a coordinate registration weighting factor, and a three-dimensional point cloud generation module for performing three-dimensional coordinate transformation of the three-dimensional point cloud coordinate system for a three-dimensional point cloud system for a three-dimensional point system for a toilet, and a three-dimensional point system for a toilet system for a three-dimensional point system for a toilet system.

Here, it will be understood by those skilled in the art that the specific operations of the respective steps in the above-described depth camera and visual large model based toilet recognition system have been described in detail in the above description of the depth camera and visual large model based toilet recognition method with reference to fig. 1 to 4, and thus, repetitive descriptions thereof will be omitted.

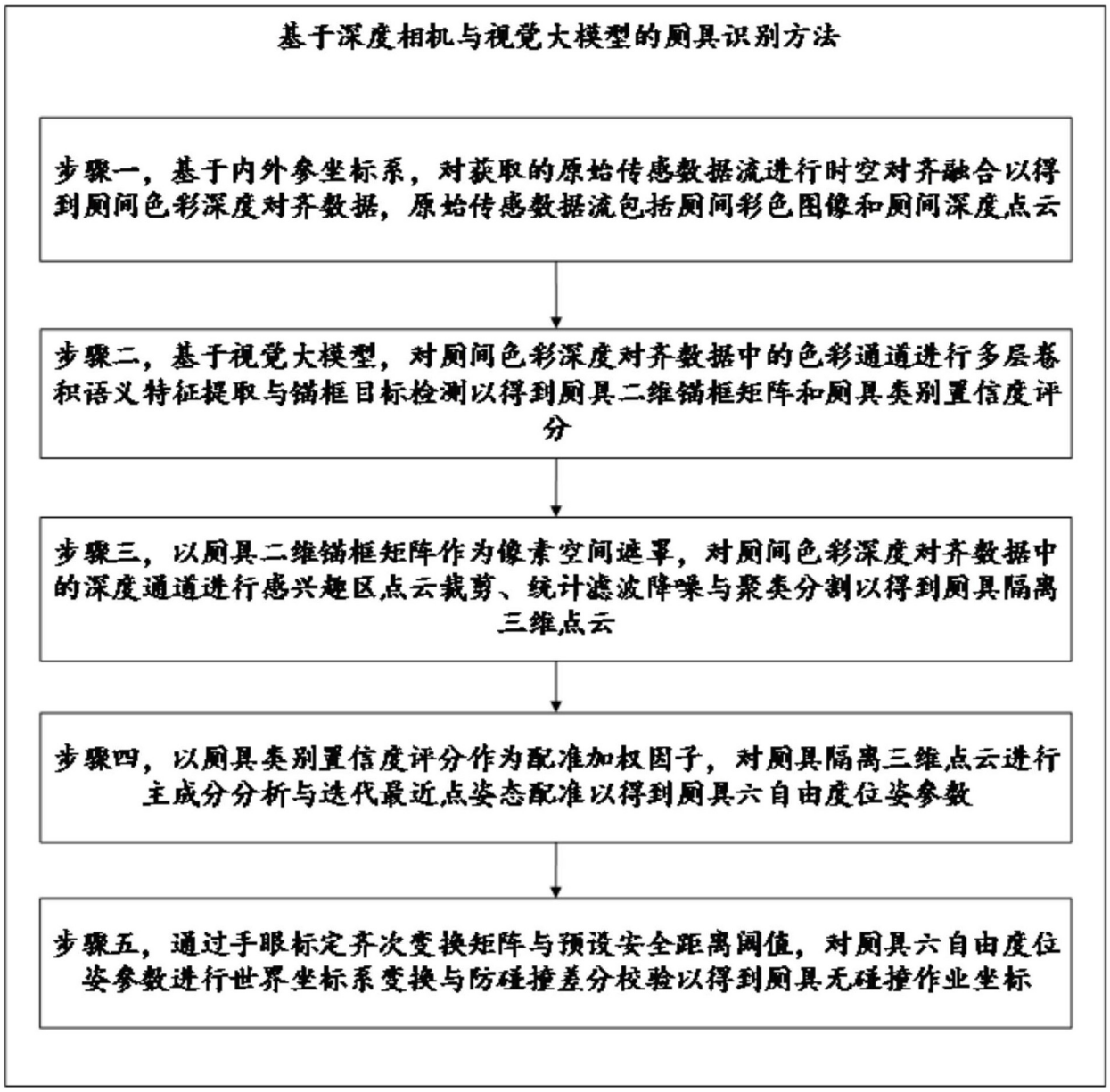

基于深度相机与视觉大模型的厕具识别方法
步骤一，基于内外参坐标系，对获取的原始传感数据流进行时空对齐融合以得到厕间色彩深度对齐数据，原始传感数据流包括厕间彩色图像和厕间深度点云
步骤二，基于视觉大模型，对厕间色彩深度对齐数据中的色彩通道进行多层卷积语义特征提取与锚框目标检测以得到厕具二维锚框矩阵和厕具类别置信度评分
步骤三，以厕具二维锚框矩阵作为像素空间遮罩，对厕间色彩深度对齐数据中的深度通道进行感兴趣区点云裁剪、统计滤波降噪与聚类分割以得到厕具隔离三维点云
步骤四，以厕具类别置信度评分作为配准加权因子，对厕具隔离三维点云进行主成分分析与迭代最近点姿态配准以得到厕具六自由度位姿参数
步骤五，通过手眼标定齐次变换矩阵与预设安全距离阈值，对厕具六自由度位姿参数进行世界坐标系变换与防碰撞差分校验以得到厕具无碰撞作业坐标

图 1

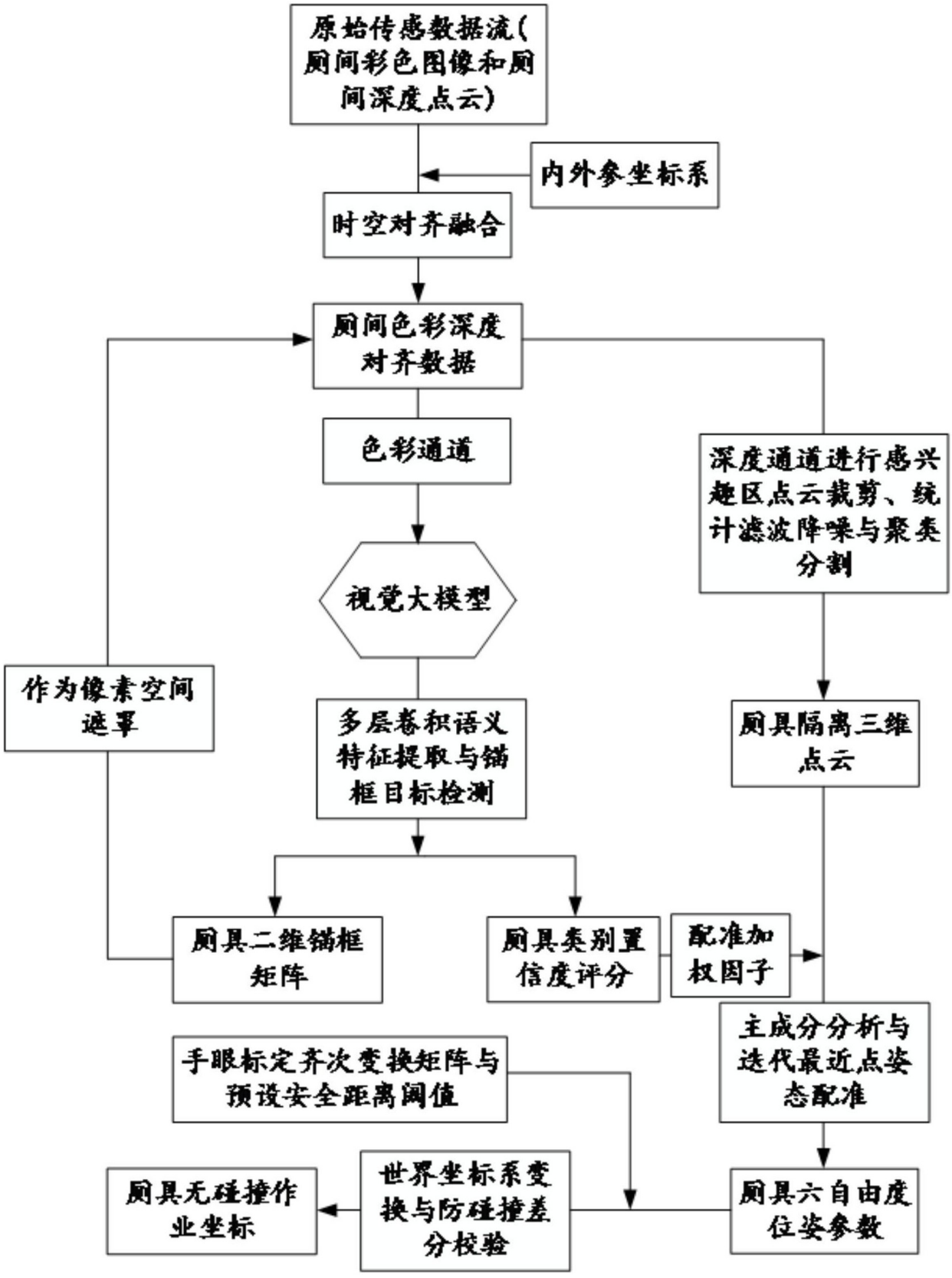

原始传感数据流(厨间彩色图像和厨间深度点云)
内外参坐标系
时空对齐融合
厨间色彩深度对齐数据
色彩通道
深度通道进行感兴趣区点云裁剪、统计滤波降噪与聚类分割
视觉大模型
作为像素空间遮罩
多层卷积语义特征提取与锚框目标检测
厨具隔离三维点云
厨具二维锚框矩阵
厨具类别置信度评分
配准加权因子
主成分分析与迭代最近点姿态配准
手眼标定齐次变换矩阵与预设安全距离阈值
厨具无碰撞作业坐标
世界坐标系变换与防碰撞差分校验
厨具六自由度位姿参数

图 2


2 视觉识别摄像头
3 磁吸式快拆接口结构
4 柔性双指夹持爪
5 六自由度串联式
可折叠机械臂
9 密闭耐压管路
6 高压喷水执行终端
7 污水回收执行终端
8 风干头执行终端
10 密封式净水箱
12 智能电控调度结构
(内置AI算法模型库
与智能排程引擎)
11 密封式污水箱
13 多模态感知数据
实时传输走线
1 深度相机
14 侧面红外传感器
15 多线激光雷达体系

图 3

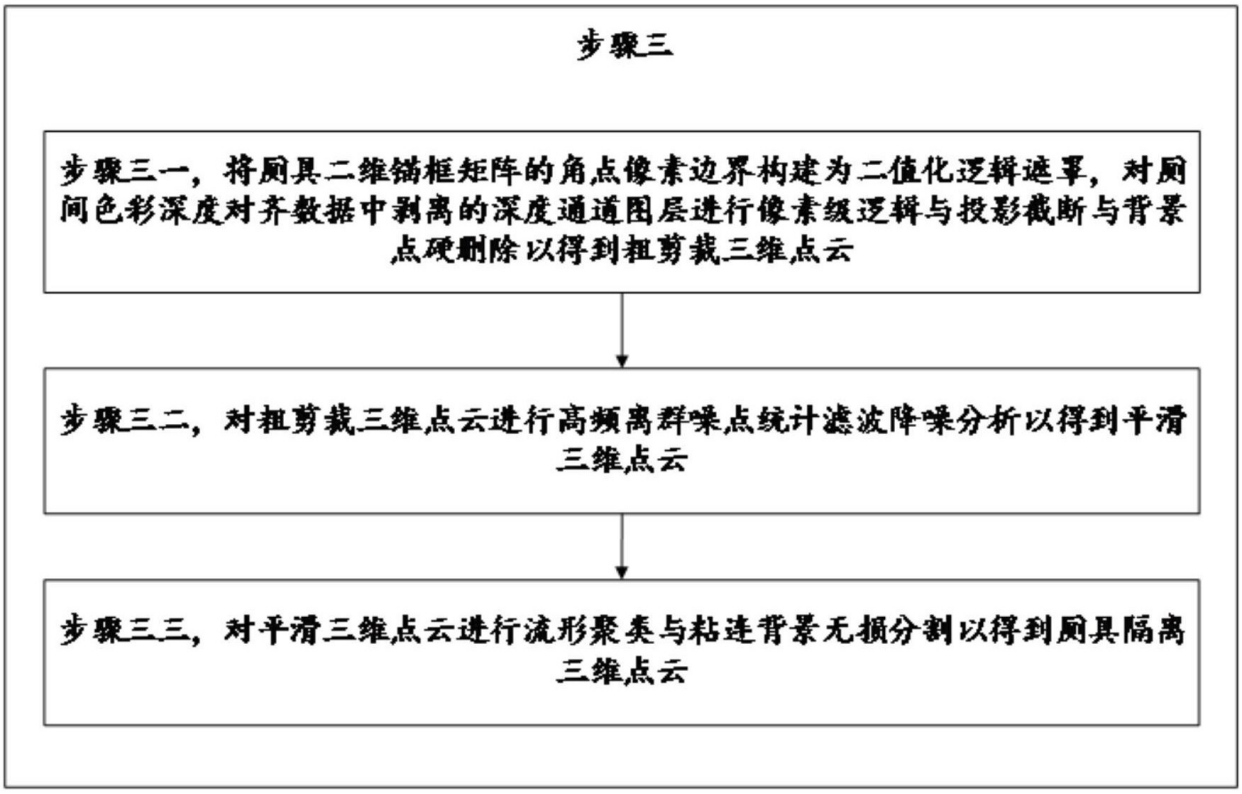

步骤三
步骤三一，将厕具二维锚框矩阵的角点像素边界构建为二值化逻辑遮罩，对厕间色彩深度对齐数据中剥离的深度通道图层进行像素级逻辑与投影截断与背景点硬删除以得到粗剪裁三维点云
步骤三二，对粗剪裁三维点云进行高频离群噪点统计滤波降噪分析以得到平滑三维点云
步骤三三，对平滑三维点云进行流形聚类与粘连背景无损分割以得到厕具隔离三维点云

图4

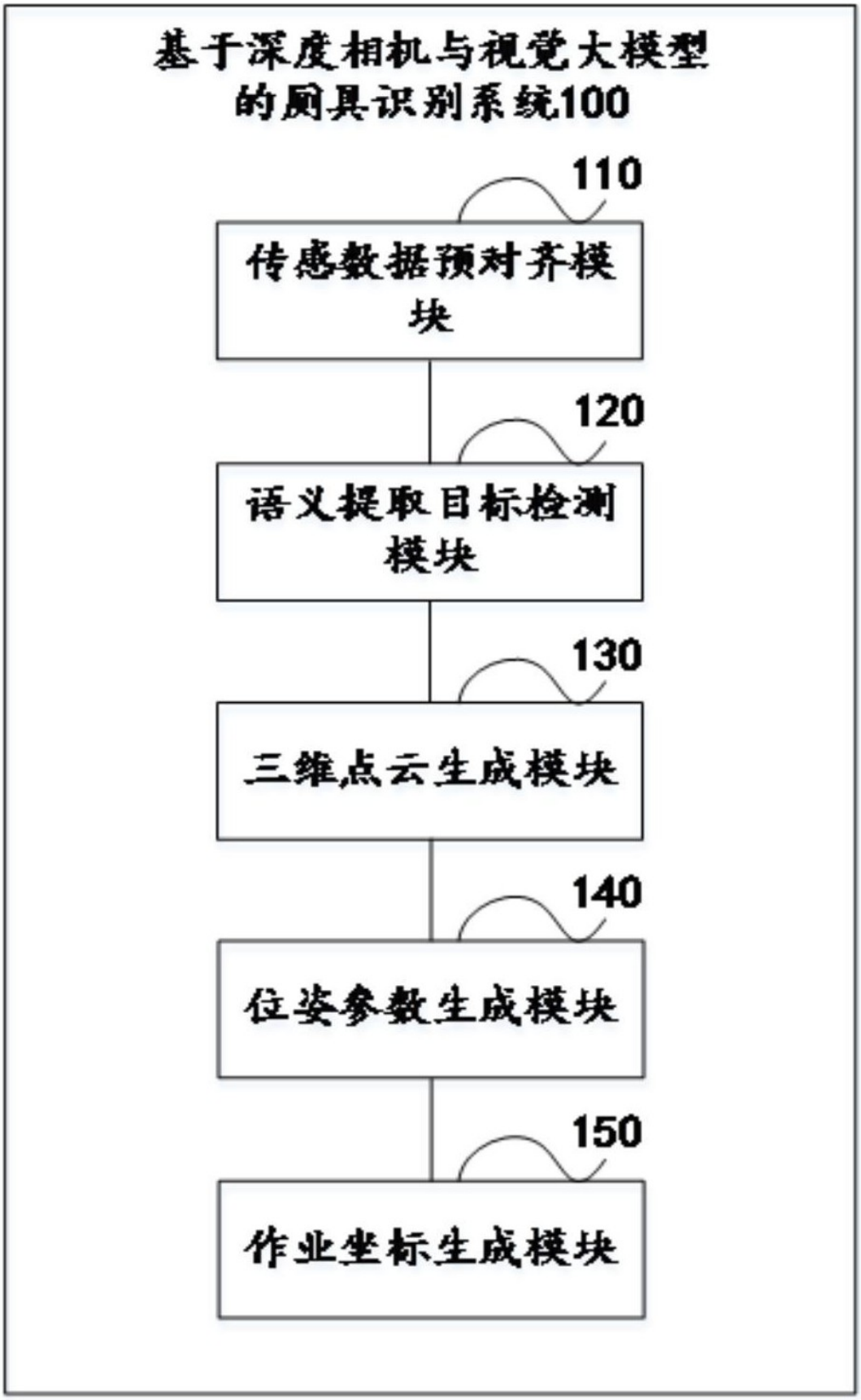

基于深度相机与视觉大模型的厨具识别系统100
110
传感数据预对齐模块
120
语义提取目标检测模块
130
三维点云生成模块
140
位姿参数生成模块
150
作业坐标生成模块

图5